Ch2 Tangent Vectors and Differentials

Germs of Functions & The Tangent Space

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The Idea: Derivations

Tangent vectors as operators

Idea. Any vector $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$ can be thought of as an *operator* on differentiable functions. If f is differentiable at p, then v assigns to f the *directional derivative*

$$v(f) = \sum_{i=1}^{n} v_i \, \frac{\partial f}{\partial r_i}(p), \qquad \text{which is a real value.}$$

It satisfies

1. **linearity** $\quad v(\lambda f + \mu g) = \lambda \, v(f) + \mu \, v(g)$;
2. **Leibniz rule** $\quad v(f \cdot g) = f(p) \cdot v(g) + v(f) \cdot g(p)$.

Why germs of functions?

The operation of taking derivatives only depends on the *local* properties of functions. This is captured by the concept of *germs of functions*.

Remark

A vector v is thus "defined" as a linear function on differentiable functions, determined by the values of $v(f)$.

Definition (Germs of functions)

Let $m \in M$. Functions f and g defined on open sets containing m are said to have the *same germ at m* if they agree on some neighborhood of m. The equivalence classes of germs of C^∞ functions are called *germs*, denoted by $\widetilde{\mathcal{F}}_m$, i.e.

$$\widetilde{\mathcal{F}}_m = \left\{ [f] \;\middle|\; f, g \in [f] \text{ if } \exists \text{ open set } U \ni m, \text{ s.t. } f\big|_U = g\big|_U \right\}.$$

The Algebra of Germs

Operations on $\widetilde{\mathcal{F}}_m$

The addition, scalar multiplication and multiplication of functions induce

$$\begin{aligned} \text{(addition)} \quad & [f] + [g] = [f + g], \\ \text{(scalar mult.)} \quad & \lambda[f] = [\lambda f], \\ \text{(multiplication)} \quad & [f] \cdot [g] = [fg]. \end{aligned}$$

Exercise

Prove that these are well-defined operations.

Remark

These operations make $\widetilde{\mathcal{F}}_m$ an *algebra* over $\mathbb{R}$. A germ $[f]$ has a well-defined value $f(m)$ at m.

Definition (The ideals F_m and F_m^k)

$$F_m = \left\{ [f] \;\middle|\; f \in \widetilde{\mathcal{F}}_m,\ f(m) = 0 \right\}.$$

Powers of the ideal F_m

F_m is an *ideal* of $\widetilde{\mathcal{F}}_m$. Let F_m^k be the k-th power of F_m: the ideal of $\widetilde{\mathcal{F}}_m$ consisting of all finite linear combinations of k-fold products of elements of F_m,

$$F_m^k = \left\{\text{linear comb. of } [f_1]\cdots[f_k],\ [f_i] \in F_m\right\}.$$

These form a descending sequence of ideals

$$\widetilde{\mathcal{F}}_m \supset F_m \supset F_m^2 \supset F_m^3 \supset \cdots$$

The Tangent Space

Tangent vectors as derivations

Definition (Tangent vector)

A *tangent vector* v at the point $m \in M$ is a *linear derivation* of the algebra $\widetilde{\mathcal{F}}_m$, i.e. for $[f], [g] \in \widetilde{\mathcal{F}}_m$ and $\lambda, \mu \in \mathbb{R}$:

(a) $v(\lambda[f] + \mu[g]) = \lambda\, v([f]) + \mu\, v([g])$;

(b) $v([f] \cdot [g]) = v([f])\, g(m) + f(m)\, v([g])$.

Let $T_m M$ or M_m denote the set of all linear derivations at m, called the *tangent space to M at m*.

Note

If $M_m \neq \varnothing$, then M_m is a *linear space*, since for $v, w \in M_m$, $\lambda, \mu \in \mathbb{R}$,

$$(\lambda v + \mu w)([f]) \triangleq \lambda\, v([f]) + \mu\, w([f]).$$

Two remarks on the definition

1. This definition of tangent spaces only works for the C^∞ case;
2. It is *intrinsic* — it does not depend on the local coordinate system.

Note (constants are killed)

For a derivation $w \in M_m$ and $[c]$ the germ of a locally constant function, $w([c]) = 0$.

Proof.

It suffices to show $w([1]) = 0$:

$$w([1]) = w([1] \cdot [1]) = w([1]) \cdot 1 + 1 \cdot w([1]) = 2\, w([1]). \qquad \square$$

$$M_m \cong (F_m / F_m^2)^*$$

The key isomorphism

Lemma

M_m is isomorphic to $(F_m/F_m^2)^*$.

Proof.

Consider $v \in M_m$. Then v is a linear function on F_m by $v : [f] \mapsto v([f])$. This function *vanishes on* F_m^2 by the Leibniz rule of v: $\forall\, [f], [g] \in F_m$,

$$v([f][g]) = v([f]) \cdot g(m) + f(m) \cdot v([g]) = 0.$$

So v can be thought of as a linear function defined on the cosets $[f] + F_m^2$, therefore $v \in (F_m/F_m^2)^*$. □

Converse direction

Conversely, for $\ell \in (F_m/F_m^2)^*$, we can define $v_\ell \in M_m$ by

$$v_\ell([f]) \triangleq \ell\big([f] - [f(m)] + F_m^2\big).$$

Then one verifies that v_ℓ is a derivation on $\widetilde{\mathcal{F}}_m$:

1. **Linearity.**
   $v_\ell\big(\lambda[f] + \mu[g]\big) = \ell\big(\lambda[f] - \lambda[f(m)] + F_m^2\big) + \ell\big(\mu[g] - \mu[g(m)] + F_m^2\big) =$
   $\lambda\, v_\ell([f]) + \mu\, v_\ell([g])$.
2. **Leibniz.** $v_\ell\big([f][g]\big) = \ell\big([f][g] - [f(m)][g(m)] + F_m^2\big)$, where

   $$[f][g] - [f(m)][g(m)] = \underbrace{\big([f] - [f(m)]\big)\big([g] - [g(m)]\big)}_{\in F_m^2} + [f(m)]\big([g] - [g(m)]\big) + \big([f] - [f(m)]\big)[g(m)],$$

   so $v_\ell([f][g]) = f(m)\, v_\ell([g]) + v_\ell([f])\, g(m)$.

An exercise hidden in the proof

Exercise

Prove that for $\ell \in (F_m/F_m^2)^*$,

$$\ell\big([f(m)] \cdot \big([g] - [g(m)]\big)\big) = f(m)\,\ell\big([g] - [g(m)]\big).$$

Theorem

$$\dim\left(F_m/F_m^2\right) = \dim M.$$

Proof of $\dim(F_m/F_m^2) = \dim M$

A Taylor lemma with integral remainder

Lemma

If $g \in C^k(U)$ for $k \geq 2$ and U is a convex open set in $\mathbb{R}^d$, then $\forall\, p, q \in U$,

$$g(q) = g(p) + \sum_{i=1}^{d} \left.\frac{\partial g}{\partial r_i}\right|_p \big(r_i(q) - r_i(p)\big) + \sum_{i,j} \big(r_i(q) - r_i(p)\big)\big(r_j(q) - r_j(p)\big) \int_0^1 (1-t)\, \left.\frac{\partial^2 g}{\partial r_j \partial r_i}\right|_{p+t(q-p)} dt.$$

Proof.

Since U is convex, for any $p, q \in U$ the segment $\{p + t(q-p) \mid t \in [0,1]\} \subset U$. Writing $p_i \triangleq r_i(p)$, $q_i \triangleq r_i(q)$, we get

$$g(q) - g(p) = \int_0^1 \frac{d}{dt}\, g\big(p + t(q-p)\big)\, dt = \sum_i \int_0^1 \left.\frac{\partial g}{\partial r_i}\right|_{p+t(q-p)} (q_i - p_i)\, dt. \quad \square$$

Proof of the Taylor lemma (continued)

For each integrand, expand around p:

$$\left.\frac{\partial g}{\partial r_i}\right|_{p+t(q-p)} = \left.\frac{\partial g}{\partial r_i}\right|_{p} + \sum_j \int_0^t \frac{\partial^2 g}{\partial r_j \partial r_i}\left(p + s(q-p)\right)(q_j - p_j)\, ds.$$

So, interchanging the order of integration ($s \leftrightarrow t$),

$$g(q) = g(p) + \sum_i \left.\frac{\partial g}{\partial r_i}\right|_p (q_i - p_i) + \sum_{i,j} (q_i - p_i)(q_j - p_j) \int_0^1 (1-t) \left.\frac{\partial^2 g}{\partial r_j \partial r_i}\right|_{p+t(q-p)} dt.$$

Proof of the theorem: spanning set

Let (U, φ) be any coordinate system about $m \in M$, with coordinate functions $(x_1, \ldots, x_d)$, $\mathbf{x}(p) = (u_1, \cdots, u_d)$, $d = \dim M$. Let $[f] \in \widetilde{\mathcal{F}}_m$. Applying $f \circ \varphi^{-1}$ to the lemma:

$$f \circ \varphi^{-1}(\mathbf{u}) - f \circ \varphi^{-1}(\mathbf{u}_0) = \sum_{i=1}^{d} \left.\frac{\partial (f \circ \varphi^{-1})}{\partial u_i}\right|_{\mathbf{u}_0} (u_i - u_i^0)$$
$$+ \sum_{i,j} (u_i - u_i^0)(u_j - u_j^0) \int_0^1 (1-t) \left.\frac{\partial^2 (f \circ \varphi^{-1})}{\partial u_j \partial u_i}\right|_{\mathbf{u}_0 + t(\mathbf{u} - \mathbf{u}_0)} dt.$$

Since $\mathbf{x}(p) := \varphi(p) = \mathbf{u}$, $x_0 = \varphi(m)$, composing with φ gives, *modulo* F_m^2,

$$[f(p)] - [f(m)] = \sum_{i=1}^{d} \left.\frac{\partial (f \circ \varphi^{-1})}{\partial u_i}\right|_{\varphi(m)} \big(x_i(p) - x_i(m)\big).$$

Hence $\big\{[x_i] - [x_i(m)]\big\}$ *spans* F_m / F_m^2.

Proof of the theorem: linear independence

We claim they are also *linearly independent*. Suppose that

$$\sum_i a_i \big(x_i - x_i(m)\big) \in F_m^2 \iff \sum_i a_i \big(u_i - u_i^0\big) \in F_{\varphi(m)}^2.$$

Then applying $\left.\dfrac{\partial}{\partial u_j}\right|_{\varphi(m)}$ to both sides gives

$$a_j = 0 \qquad \forall\, j.$$

Therefore $\big\{[x_i] - [x_i(m)]\big\}_{i=1}^d$ is a basis of F_m/F_m^2, and

$$\dim\big(F_m/F_m^2\big) = d = \dim M. \quad \square$$

The Usual View: Coordinate Basis

Tangent vectors acting on functions

In practice, a tangent vector operates on *functions* rather than on their germs. Let f be a differentiable function defined on a neighborhood of $p \in M$. For $v \in T_pM$ (or M_p), we define

$$v(f) \triangleq v([f]).$$

So $v(f) = v(g)$ whenever f and g agree on a neighborhood of p.

$$v(\lambda f + \mu g) = \lambda\, v(f) + \mu\, v(g) \tag{a}$$

$$v(f \cdot g) = v(f)\, g(p) + f(p)\, v(g) \tag{b}$$

Coordinate tangent vectors

Definition (Coordinate tangent vectors)

Let (U, φ) be a coordinate system with coordinate functions $x_1, \ldots, x_d$, and let $p \in U$. For $i = 1, \ldots, d$ define

$$\left.\frac{\partial}{\partial x_i}\right|_p (f) \triangleq \left.\frac{\partial (f \circ \varphi^{-1})}{\partial r_i}\right|_{\varphi(p)}$$

for each C^∞ function f on a neighborhood of p. We also use the notation $\left.\frac{\partial f}{\partial x_i}\right|_p = \left.\frac{\partial}{\partial x_i}\right|_p (f)$.

Remark (a)

Clearly $\left.\frac{\partial}{\partial x_i}\right|_p (f)$ depends only on the germ of f at p, and (a), (b) are satisfied. So $\left.\frac{\partial}{\partial x_i}\right|_p$ is a tangent vector at p.

The coordinate basis

Moreover $\left\{ \frac{\partial}{\partial x_i}\Big|_p \right\}_{i=1}^{d}$ is a basis of T_pM — indeed the basis *dual* to $\{[x_i - x_i(p)]\}$ of F_p/F_p^2, since

$$\frac{\partial}{\partial x_i}\Big|_p \big(x_j - x_j(p)\big) = \delta_{ij}.$$

(b) If $w \in T_pM$, then

$$w = \sum_{i=1}^{d} w(x_i)\ \frac{\partial}{\partial x_i}\Big|_p ,$$

since both sides give the same result when acting on the functions $x_i - x_i(p)$.

Change of coordinates

(c) If (U, φ) and (V, ψ) are coordinate systems about p with coordinate functions $(x_1, \ldots, x_d)$ and $(y_1, \ldots, y_d)$, then by (b)

$$\left.\frac{\partial}{\partial y_j}\right|_p = \sum_{i=1}^{d} \left.\frac{\partial x_i}{\partial y_j}\right|_p \left.\frac{\partial}{\partial x_i}\right|_p .$$

Note $\left(\frac{\partial}{\partial x_i}\right)$ depends on φ, not only on x_i; so $x_i = y_i$ does *not* necessarily imply $\frac{\partial}{\partial x_i} = \frac{\partial}{\partial y_i}$.

(d) On the Euclidean space $\mathbb{R}^d$, the tangent vectors are all the linear combinations of $\frac{\partial}{\partial r_i}$.

The C^k caveat

Note

The proof that $\mathbf{dim}(F_m/F_m^2) = \mathbf{dim}\, M$ fails in the C^k case ($k < \infty$): the remainder term in the Taylor lemma is only C^{k-1}, and F_m/F_m^2 is in fact always infinite-dimensional in the C^k case.

There are various ways to define tangent vectors in the C^k case so that $\mathbf{dim}\, T_pM = \mathbf{dim}\, M$. For example, one may define a tangent vector at p as a mapping assigning to each function (C^k locally about p) a real number such that, whenever (U, φ) is a coordinate system on a neighborhood of p, there exists a list of numbers $(a_1, \ldots, a_d)$ with

$$v(f) = \sum_{i=1}^{d} a_i \left.\frac{\partial(f \circ \varphi^{-1})}{\partial r_i}\right|_{\varphi(p)}.$$

The Differential

The differential of a map

Definition (The differential)

Let $\Psi : M \to N$ be a C^∞ map and $p \in M$. The *differential* (or tangent map) of Ψ at p is the linear map

$$\Psi_{*,p} : T_pM \longrightarrow T_{\Psi(p)}N$$

defined as follows: for $w \in T_pM$ and a C^∞ function f on a neighborhood of $\Psi(p)$,

$$\Psi_{*,p}(w)(f) \triangleq w(f \circ \Psi) = d\Psi_p(w)(f).$$

Ψ is called *non-singular* at p if $\ker(\Psi_{*,p}) = \{0\}$.

The dual map $\Psi_p^* : (T_{\Psi(p)}N)^* \longrightarrow (T_pM)^*$ is defined by $\Psi_p^*(\omega)(w) = \omega(\Psi_*(w)) = \omega(d\Psi(w))$.

Example: the differential of a function

If $f : M \to \mathbb{R}$ is smooth, $w \in T_pM$ and $f(p) = r_0$, then

$$df(w) = w(f) \left.\frac{d}{dr}\right|_{r_0}.$$

Proof.

If $w = \sum_i v_i \left.\frac{\partial}{\partial x_i}\right|_p$, then $df(w) = \sum_i v_i \, df\left(\frac{\partial}{\partial x_i}\right)$. For any C^∞ function g about r_0,

$$df(w)(g) = w(g \circ f) = \sum_i v_i \frac{\partial g}{\partial r} \frac{\partial f}{\partial x_i} = w(f) \frac{d}{dr}(g). \qquad \square$$

In this case $df \in (T_pM)^*$ is simply given by $df(w) \triangleq w(f)$.

Jacobian matrix and dual basis

(a) Let $(U, x_1, \ldots, x_d)$ and $(V, y_1, \ldots, y_d)$ be coordinate systems about p and $\Psi(p)$. Then

$$d\Psi\left(\left.\frac{\partial}{\partial x_j}\right|_p\right) = \sum_{i=1}^{d} \left.\frac{\partial(y_i \circ \Psi)}{\partial x_j}\right|_p \left.\frac{\partial}{\partial y_i}\right|_{\Psi(p)}.$$

The matrix $\left(\left.\frac{\partial(y_i \circ \Psi)}{\partial x_j}\right|_p\right)$ is called the *Jacobian* of Ψ.

(b) If $(U, x_1, \ldots, x_d)$ is a coordinate system on M and $p \in U$, then $\left\{(dx_i)_p\right\}$ is the basis of $(T_pM)^*$ dual to $\left\{\left.\frac{\partial}{\partial x_i}\right|_p\right\}$. For $f : M \to \mathbb{R}$ smooth,

$$df_p = \sum_{i=1}^{d} \left.\frac{\partial f}{\partial x_i}\right|_p (dx_i)_p.$$

Chain rule and pullback

(c) **Chain rule.** Let $\Psi : M \to N$ and $\Phi : N \to X$ be C^∞ maps. Then

$$d(\Phi \circ \Psi)_p = d\Phi\big|_{\Psi(p)} \circ d\Psi\big|_p.$$

(d) If $\Psi : M \to N$ and $f : N \to \mathbb{R}$ are C^∞, then

$$\Psi^*(df)_{\Psi(p)} = d(f \circ \Psi)_p,$$

because for all $w \in T_pM$,

$$\Psi^*(df)_{\Psi(p)}(w) = d(f \circ \Psi)\big|_p(w).$$

Smooth Curves

Smooth curves

Definition (Smooth curve)

A C^∞ map $\sigma : (a, b) \to M$ is called a *smooth curve* in M. Its tangent vector at $t \in (a, b)$ is

$$\dot{\sigma}(t) \triangleq d\sigma\left(\left.\frac{d}{dt}\right|_t\right) = \sigma_*\left(\left.\frac{d}{dt}\right|_t\right) \in M_{\sigma(t)},$$

given in coordinates by $\dot{\sigma}(t) = \sum_{i=1}^{d} \left.\frac{\partial x_i}{\partial t}\right|_t \left.\frac{\partial}{\partial x_i}\right|_{\sigma(t)}$.

Remark

Every $w \in T_pM$ is the tangent vector to a smooth curve: choose (U, φ) centered at p with $w = d\varphi^{-1}(\left.\frac{\partial}{\partial r_1}\right|_0)$; then w is tangent at 0 to $t \mapsto \varphi^{-1}(t, 0, \ldots, 0)$.

Remarks on curves

- Two curves σ, τ with $\sigma(t_0) = \tau(t_0) = p$ have the same tangent vector at t_0 iff $\left.\frac{d(f\circ\sigma)}{dt}\right|_{t_0} = \left.\frac{d(f\circ\tau)}{dt}\right|_{t_0}$ for all f which are C^∞ about p.
- In $\mathbb{R}^d$: $\dot{\sigma}(t) = \sum_i \left.\frac{d\sigma_i}{dt}\right|_t \left.\frac{\partial}{\partial r_i}\right|_{\sigma(t)} \equiv (\dot{\sigma}_1, \ldots, \dot{\sigma}_d)$.

An Application

A map with vanishing differential is constant

Theorem

Let Ψ be a C^∞ mapping of the connected manifold M into N. If $d\Psi|_p \equiv 0$ for all $p \in M$, then Ψ is a constant map.

Proof.

Let $q \in \Psi(M)$. The set $\Psi^{-1}(\{q\})$ is closed; we show it is open. Let $m \in \Psi^{-1}(\{q\})$, and choose coordinate systems $(U, x_1, \ldots, x_d)$, $(V, y_1, \ldots, y_e)$ about m and q with $\Psi(U) \subset V$. Then for all $p \in U$,

$$0 = d\Psi\left(\frac{\partial}{\partial x_j}\right) = \sum_{i=1}^{e} \left.\frac{\partial(y_i \circ \Psi)}{\partial x_j}\right|_p \left.\frac{\partial}{\partial y_i}\right|_{\Psi(p)} \implies \left.\frac{\partial(y_i \circ \Psi)}{\partial x_j}\right|_p = 0 \quad \forall\, i, j.$$

So the $y_i \circ \Psi$ are constant on U, i.e. $\Psi(U) = \{q\}$, and $\Psi^{-1}(\{q\})$ is open. $\square$

Tangent and Cotangent Bundles

Tangent and cotangent bundles as sets

Let M be a C^∞ manifold with differentiable structure $\mathcal{F}$.

Definition (Tangent & cotangent bundle)

$$TM = \bigcup_{p \in M} T_pM, \qquad T^*M = \bigcup_{p \in M} T_p^*M$$

as two sets of tangent / cotangent vectors.

Definition (Natural projection)

$$\pi : TM \longrightarrow M, \qquad \pi^* : T^*M \longrightarrow M,$$

sending v (at p) $\mapsto p$ and ω (at p) $\mapsto p$, where $v \in T_pM$, $\omega \in T_p^*M$.

Local trivializations

Definition (Local trivialization)

Let (U, φ) be a local coordinate system in $\mathcal{F}$, with coordinate functions $x_1, \ldots, x_d$. Define maps

$$\widetilde{\varphi} : \pi^{-1}(U) \longrightarrow \mathbb{R}^{2d}, \qquad \widetilde{\varphi}^* : (\pi^*)^{-1}(U) \longrightarrow \mathbb{R}^{2d}$$

by

$$\widetilde{\varphi}(w) = \big(\varphi(p),\ a_1, \ldots, a_d\big),$$

if $w \in T_pM$ can be written as $w = \sum_{i=1}^{d} a_i \left.\frac{\partial}{\partial x_i}\right|_p \in T_pM \subset \pi^{-1}(U)$; and similarly $\widetilde{\varphi}^*(\omega) = \big(\varphi(p),\ b_1, \ldots, b_d\big)$ if $\omega = \sum_{j=1}^{d} b_j\, dx_j \in T_p^*M$.

Note

Both $\widetilde{\varphi}$ and $\widetilde{\varphi}^*$ are one-to-one maps onto open subsets $\varphi(U) \times \mathbb{R}^d \subset \mathbb{R}^{2d}$.

The topology on TM

Outline of the construction of the C^∞ manifold structure on TM.

1. TM is a topological space, which is Hausdorff, second countable and locally compact.

   - The collection

   $$\mathcal{B} \triangleq \left\{ \widetilde{\varphi}^{-1}(W) \;\middle|\; W \subseteq \mathbb{R}^{2d} \text{ open},\ W \neq \varnothing,\ \forall\, (U, \varphi) \in \mathcal{F} \right\}$$

   forms a basis for a topology on TM.

   - The topology $\mathcal{V}$ generated by $\mathcal{B}$ makes $(TM, \mathcal{V})$ Hausdorff, second countable and locally compact.
   - TM is a local Euclidean space because of the local trivialization $\widetilde{\varphi}$ (which provides the homeomorphism).

The differentiable structure on TM

The compatible differentiable structure $\widetilde{\mathcal{F}}$ of TM is the maximal collection of local coordinate systems containing

$$\widetilde{\mathcal{F}} \supset \left\{ \left(\pi^{-1}(U), \widetilde{\varphi}\right) \mid (U, \varphi) \in \mathcal{F} \right\}.$$

Exercise

It is necessary to show that, given two local coordinate systems $\left(\pi^{-1}(U), \widetilde{\varphi}\right)$ and $\left(\pi^{-1}(V), \widetilde{\psi}\right)$, the transition map

$$\widetilde{\psi} \circ \widetilde{\varphi}^{-1}$$

is a C^∞ map. (Check it!)

TM and T^*M with these differentiable structures are called the *tangent bundle* and *cotangent bundle*, respectively. It is sometimes convenient to write points of TM as pairs (p, v), where $p \in M$ and $v \in T_pM$.

Differentials of bundle maps

If $\Psi : M \to N$ is a C^∞ map, then the differential of Ψ defines a mapping of the tangent bundles

$$d\Psi = \Psi_* : TM \longrightarrow TN,$$

where

$$d\Psi(p, w) = \big(\Psi(p),\ d\Psi(w)\big).$$

Exercise

Check that this is a C^∞ map.